A Comparative Study of Frozen Visual Representations for Seven-Class Dermoscopic Lesion Classification

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Abstract—This report documents the architecture, experimental results, and implementation status of a research prototype for seven-class dermoscopic lesion classification. The project compares frozen ImageNet EfficientNet-B0 features and frozen PanDerm vision-transformer representations with linear, multilayer perceptron (MLP), gradient-boosted tree, and one-versus-rest specialist classifiers. A separate federated-learning path implements Flower, FedAvg/FedProx options, and optional Opacus differential privacy. On the official ISIC 2018 Task 3 test cohort, PanDerm Base embeddings with an MLP achieved 80.56% accuracy and 70.30% macro-F1; a local reproduction of the PanDerm Large authors' linear-probe recipe achieved 82.94% accuracy and 71.58% macro-F1. These are historical, repeatedly inspected benchmark results, not fresh blind validation. Improved EfficientNet features achieved 61.46% macro-F1. A later PanDerm adaptation used both former test cohorts for additional fitting, so those scores are fitting diagnostics only. The federated path was demonstrated only with synthetic data and did not establish lesion-classification utility. The prototype is not intended for clinical use.

Index Terms—dermoscopy, skin lesion classification, transfer learning, vision transformer, EfficientNet, PanDerm, class imbalance, federated learning.

I. INTRODUCTION

This project studies image classification over seven diagnoses: actinic keratoses (akiec), basal cell carcinoma (bcc), benign keratosis-like lesions (bkl), dermatofibroma (df), melanoma (mel), melanocytic nevus (nv), and vascular lesion (vasc). Its central question is how well fixed pretrained visual representations support these classes using smaller task-specific classifiers.

The codebase grew from a federated CNN prototype into a local research workbench. Evidence spans independent experiment families rather than one matched leaderboard. The report keeps frozen-feature studies separate from the exploratory federated simulation and identifies the protocol alongside each result.

The work provides a reproducible embedding and classification pipeline; comparisons among linear, nonlinear, tree, and specialist heads; a verified local PanDerm encoder and authors' probe reference; and a Flower/Opacus prototype whose measured federated evidence is limited to synthetic smoke data.

II. RELATED WORK

HAM10000 is a dermatoscopic image collection with seven diagnosis categories [1]. ISIC 2018 Task 3 provides a lesion-diagnosis evaluation setting [2]. PanDerm is a multimodal dermatology foundation model pretrained across clinical imaging modalities and institutions; this project uses released image encoders as fixed representation extractors [3]. EfficientNet is used as an ImageNet-pretrained convolutional feature extractor [4].

III. SYSTEM ARCHITECTURE

A. Image Classification Flow

The inference path is: dermoscopic image → RGB decoding and deterministic preprocessing → frozen encoder → feature vector → classifier head → seven class scores and predicted label. The application loads saved model artifacts and displays predictions and results. Training and inference code paths are separate; artifacts retain parameters, configuration, metrics, and image-level predictions.

Input	Preprocess	Frozen encoder	Embedding	Classifier	Output
Image	Resize/crop normalize	EfficientNet or PanDerm	1280 / 768 / 1024 values	Linear / MLP / XGBoost	Seven scores and class

Fig. 1. Centralized classification path. Federated training is a separate experimental branch.

B. Representation Backbones

EfficientNet-B0 uses ImageNet-pretrained weights in evaluation mode without gradient updates and emits 1,280-dimensional descriptors. In the improved multi-view run, a canonical crop descriptor is concatenated with the mean of eight square-symmetry full-image view descriptors, yielding 2,560 dimensions. Cached features support classifier trials without repeatedly running the encoder.

PanDerm Base and Large emit 768- and 1,024-dimensional embeddings. The Large encoder has 303,326,208 parameters and remains frozen in local head comparisons. Preprocessing follows the pinned upstream evaluation transform: RGB, resize to 256 pixels, center crop to 224 pixels, and channel normalization. Encoder source and checkpoint tensors are hash-checked and loaded strictly. The measured system is encoder plus classifier, not a bare zero-shot classifier.

C. Classifier Heads

Linear probe: multinomial logistic regression. Custom runs use training-fitted scaling and class weighting; the PanDerm authors' reference preserves the released raw-feature probe with C=71.68. MLP: LayerNorm, ReLU, dropout, and seven output logits. Representative shapes are 1280→256→64→7, 1024→256→64→7, and 1024→192→64→7, trained with AdamW and validation early stopping. XGBoost: shallow boosted trees over frozen features, with class weights and validation stopping; one PanDerm experiment used supervised six-dimensional LDA before the trees. Lesion specialists: seven independently selected one-versus-rest classifiers with fold-derived calibration and thresholds; their binary metrics differ from final seven-way argmax metrics.

Scaling, projection, and model selection are fitted within training partitions or folds. Validation selects settings as declared for each experiment. Validation and test cohorts have nevertheless been examined repeatedly across project experiments.

D. Federated Branch

Flower implements client-local updates, IID or Dirichlet label partitions, FedAvg, and optional FedProx. The default configuration specifies five clients and Dirichlet α=0.5. Opacus can add per-example gradient clipping, noise, and privacy accounting. Raw images remain local in the simulation. These are implementation options, not evidence of a deployed multi-hospital system or certified privacy. The saved demonstration uses two clients, two rounds, IID synthetic data, and differential privacy disabled.

IV. EXPERIMENTAL PROTOCOL

A. Evaluation Populations and Metrics

The improved EfficientNet experiment documents 7,014 training, 1,509 internal validation, 1,492 internal test, 193 official validation, and 1,512 official test images. Training, validation, and internal test partitions are lesion-disjoint. The separate PanDerm released HAM split is 8,207 training, 575 validation, and 1,232 test images; these counts must not be combined with the EfficientNet split. External grouping identifiers do not support cross-release lesion or patient-independence claims.

Accuracy is exact-label accuracy. Macro-F1 weights classes equally; weighted-F1 weights by class support; balanced accuracy averages class recall; macro-F0.5 emphasizes precision over recall. Macro metrics expose minority-class performance that accuracy can hide.

B. Model Selection

Improved EfficientNet selects representation and head settings using internal validation macro-F1, then freezes candidates before test scoring. PanDerm custom heads use either released validation or training-fold cross-validation as specified in their experiment. The authors' probe is fixed. Since evaluation cohorts have been reused across the project history, these results are historical comparisons rather than prospective blind estimates.

V. RESULTS

A. Improved EfficientNet Features

Table I reports the improved frozen EfficientNet comparison. The linear classifier was selected by internal validation macro-F1 and achieved 0.6146 macro-F1 on the official test cohort. XGBoost had the strongest macro-F0.5 among these heads, so no family dominates every measure.

Head	Val. F1	Internal F1	Official F1	Official F0.5
Linear	0.6105	0.5976	0.6146	0.6150
MLP	0.6078	0.6083	0.6068	0.6105
XGBoost	0.5655	0.6020	0.5866	0.6171

TABLE I. IMPROVED EFFICIENTNET RESULTS (MACRO METRICS).

B. PanDerm Comparisons

Table II summarizes separate frozen PanDerm experiments on the official 1,512-image cohort. Base+MLP was validation-selected among Base heads. Large results used a different fitting split and selection protocol; they are not an architecture-only ablation. All figures are historical comparisons.

Encoder/head	Selection	Accuracy	Macro-F1	Macro-F0.5
Base + MLP	Validation	0.8056	0.7030	0.7055
Base + linear	Validation	0.7917	0.6773	0.6865
Base + XGBoost	Validation	0.7566	0.6408	0.6558
Large + author probe	Fixed recipe	0.8294	0.7158	0.7415
Large + custom linear	Validation	0.8313	0.7380	0.7463
Large + MLP (192-64)	Validation	0.8214	0.7096	0.7202
Large + XGBoost+LDA	Validation	0.8201	0.7267	0.7235

TABLE II. PANDERM OFFICIAL-TEST METRICS (FRACTIONS).

On the distinct 1,232-image released HAM test split, the Large authors' probe reached 92.37% accuracy, 80.98% macro-F1, and 92.22% weighted-F1. The Large 192-64 MLP reached 92.13% accuracy and 81.73% macro-F1, with 98.22% training macro-F1. The fit-to-test gap warrants caution; these results are not interchangeable with the official ISIC cohort.

C. Specialists and Regularization

Regularized specialists achieved 0.5998 macro-F1 and 0.6453 macro-F0.5 on official test. The SMOTE counterpart reached 0.5858 and 0.6478. The improved EfficientNet linear model scored 0.6146 macro-F1 on the same repeatedly evaluated cohort. Specialization therefore did not improve overall macro-F1 over that reference. It represents a different per-class operating strategy, not a guaranteed overall gain.

D. Adaptation Scores

The latest joint refinement applied eight further fitting updates to both former test cohorts (HAM n=1,232; official ISIC n=1,512). PanDerm Large 192-64 then scored 93.99% accuracy/87.38% macro-F1 on HAM and 85.91%/79.02% on ISIC. These are fitting scores, not test metrics. HAM validation was excluded from updates but had been inspected previously, so it is not fresh independent confirmation.

E. Federated Demonstration

The saved two-client, two-round synthetic FedAvg run used one local epoch and no DP. Final accuracy was 14.29%, nominal chance for seven balanced categories. This verifies pipeline execution only. No real-image FL performance, privacy-utility result, or matched centralized comparison is established by this receipt.

VI. DISCUSSION

Frozen pretrained representations support useful lightweight classifiers, but accuracy and weighted-F1 alone obscure minority-diagnosis errors. PanDerm results vary with backbone scale, probe design, fitting cohort, and selection procedure. The Large custom linear result is a promising historical comparison but does not isolate backbone scale causally from the Base result. The local authors' probe follows released implementation artifacts; it is not a claim of exact manuscript reproduction.

The strongest centralized scores did not come from federated or differentially private training. A real FL study needs

institution-level partitions, client-specific test results, measured privacy budgets, and a matched centralized baseline.

VII. LIMITATIONS AND RESPONSIBLE USE

The prototype has no prospective clinical validation, multi-institution evaluation, fairness assessment, or workflow study. Pretraining overlap has not been independently excluded. Repeated inspection of evaluation cohorts creates adaptive-reuse risk even if a particular run did not use test labels for selection. Cross-release patient/lesion independence is unknown. Differential privacy was disabled in the saved federated run; image-level accounting would not by itself prove patient-level protection. Decision scores are not clinical probabilities. The upload checker does not verify that an input depicts a lesion. This is not a medical device or diagnostic tool.

VIII. CONCLUSION

The project provides an auditable framework for comparing frozen visual encoders with lightweight classifiers for seven-class lesion recognition. The strongest historical official-cohort macro-F1 is 0.7380 from a validation-selected PanDerm Large linear head; PanDerm Base+MLP reached 0.7030 and improved EfficientNet linear features reached 0.6146. These are protocol-dependent, repeatedly inspected measurements, not fresh clinical evidence. Adapted-cohort scores are fitting diagnostics, and the federated system has only been smoke-tested with synthetic data. Priorities are an untouched external cohort, matched encoder/head protocols, patient-level grouping, uncertainty reporting, and real multi-institution federated evaluation.

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Prepared from repository source, experiment reports, metric receipts, and verification artifacts available October 28, 2026. Dataset discussion is limited to details needed to interpret evaluation protocols.

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